Beyond Homo Economicus: Computational Verification of Amartya Sen's Meta-Ranking Theory in Multi-Agent Social Dilemmas

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https://github.com/Yesol-Pilot/EthicaAI

Abstract

Can artificial agents develop genuine moral commitment beyond self-interest? We formalize Amartya Sen's **Meta-Ranking** theory—preferences over preferences implementing moral commitment—within a Multi-Agent Reinforcement Learning (MARL) framework. Our dynamic commitment mechanism λ_t, conditioned on resource availability and Social Value Orientation (SVO), is tested across **8 environments** with up to **1,000 agents**. Three key findings emerge: (1) Dynamic meta-ranking significantly enhances collective welfare ($p<0.001$ via LMM with agent-level random effects), while static SVO injection fails ($p=0.64$). (2) Non-significant cooperation rates mask *emergent role specialization*—a division of labor between "Cleaners" and "Eaters" that improves sustainability. (3) **Situational Commitment** achieves utilitarian-optimal welfare (147.7) using only local information, matching the performance of global utilitarianism (99.6% ESS). We further demonstrate **scale invariance** (SII$\approx$1.0 up to 1,000 agents), **Byzantine robustness** against 50% adversarial populations, and superiority over SOTA baselines (IA, SI, LOLA) in adversarial settings with $2.5\times$ resilience advantage. Code and data: `https://anonymous.4open.science/r/EthicaAI`.

1 Introduction

Rational Choice Theory defines human behavior as utility maximization under constraints. Sen [11] critiqued this as the "Rational Fool," arguing that genuine moral behavior requires *commitment*—acting against immediate self-interest—mediated through *meta-rankings*: preferences over preference orderings.

While recent work has explored value alignment in MARL [10, 2], these approaches typically inject fixed moral values as reward shaping, ignoring Sen's crucial insight that commitment must adapt to survival constraints. We bridge this gap with five contributions:

1. **(C1) Dynamic Meta-Ranking**: First MARL implementation of Sen's theory via dynamic $\lambda_t = f(\theta_i, R_t, \lambda_{t-1})$. Convergence (Thm. 1) and robust tracking (Thm. 2) are proved.
2. **(C2) Role Specialization**: Discovery of emergent division of labor that resolves the "cooperation paradox"—non-significant cooperation rates coexist with significant welfare gains.
3. **(C3) Local vs Global Optimality**: Comparison of 5 moral frameworks showing Situational Commitment matches Utilitarian optimality without requiring global state information.
4. **(C4) Robustness at Scale**: Verification across 1,000 agents, 5 network topologies, and 50% adversarial populations.
5. **(C5) SOTA Comparison**: First systematic quantitative comparison against IA, SI, and LOLA in adversarial multi-agent settings.

2 Related Work

Moral MARL. Rodríguez-Soto et al. [10] embed static moral values in MARL rewards but demonstrate limited scalability (2 agents). Calvano et al. [1] show Q-learning agents can learn collusion, motivating the need for moral mechanisms. Christoffersen et al. [2] use LLMs for moral reasoning but lack formal foundations.

Opponent Shaping. LOLA [4] differentiates through opponent parameter updates to shape their learning, achieving cooperation in 2-player games. M-FOS [8] extends this model-free via meta-learning over episode batches. However, both approaches require access to opponent parameters or rollouts — a strong assumption that limits decentralization. Meta-ranking operates purely from *internal state* (θ_i, R_t), requiring no opponent modeling.

Incentive Design. LOPT [12] learns Pigovian tax schedules via a centralized regulator to internalize externalities. Hughes et al. [5] inject inequity aversion (Fehr-Schmidt penalties) as reward shaping. Jaques et al. [7] reward agents for causally influencing others' actions. These approaches either require a *central authority* (LOPT) or operate on *static* social preferences (IA, SI). Meta-ranking is both **decentralized** and **dynamic**: λ_t adapts to resource state without external coordination, and the commitment level itself evolves (Table 4).

Evolutionary Game Theory. Nowak [9] establishes 5 mechanisms for cooperation evolution. Our contribution extends this by showing meta-ranking constitutes a novel 6th mechanism: *commitment-mediated cooperation*.

Mechanism Design. Unlike mechanism design approaches that rely on external incentives [6], meta-ranking implements *internal* moral commitment, making it robust to mechanism circumvention.

3 Method

3.1 Meta-Ranking Reward Function

Definition 1 (Meta-Ranking Agent)**.** *Agent i with SVO angle θ_i maximizes:*

$$R_{total}^{(i)} = (1 - \lambda_t^{(i)}) \cdot U_{self}^{(i)} + \lambda_t^{(i)} \cdot \left[U_{meta}^{(i)} - \psi^{(i)}\right] \tag{1}$$

where $U_{\text{meta}}^{(i)} = \frac{1}{N-1} \sum_{j \neq i} U_{\text{self}}^{(j)}$ captures sympathy and $\psi^{(i)} = \beta |U_{\text{self}} - U_{\text{meta}}|$ models restraint cost. The parameter $\lambda_t \in [0, 1]$ governs the degree of moral commitment: $\lambda_t = 0$ yields pure self-interest, while $\lambda_t = 1$ represents full altruism.

3.2 Dynamic Commitment Mechanism

The key innovation is that λ_t adapts to resource level R_t:

$$\lambda_t^{(i)} = \alpha \lambda_{t-1}^{(i)} + (1 - \alpha) \cdot g(\theta_i, R_t) \tag{2}$$

where $g(\theta, R) = \begin{cases} \max(0, \sin\theta \cdot 0.3) & R < R_{\text{crisis}} \\ \min(1, 1.5 \sin\theta) & R > R_{\text{abundance}} \\ \sin\theta \cdot (0.7 + 1.6R) & \text{otherwise} \end{cases}$

This implements Sen's insight computationally: when resources are scarce ($R < R_{\text{crisis}}$), agents reduce commitment to ensure survival; when abundant ($R > R_{\text{abundance}}$), they increase prosocial behavior. The EMA parameter α prevents oscillation.

Theorem 1 (Convergence)**.** *For $\alpha \in (0, 1)$ and bounded g, the λ_t update rule (Eq. 2) is a contraction mapping with unique fixed point $\lambda^* = g(\theta, R^*)$, converging geometrically at rate α.*

Theorem 2 (Robust Tracking)**.** *Under time-varying resources R_t, the tracking error is bounded by $|\lambda_t - \lambda^*(R_t)| \leq \rho^t |\lambda_0 - \lambda_0^*| + \frac{D}{1-\rho}$, where $\rho = (1-\alpha)$ and $D = \max_t |\lambda^*(R_{t+1}) - \lambda^*(R_t)|$.*

Proof sketch. By the Banach fixed-point theorem, the EMA operator $T(\lambda) = \alpha\lambda + (1-\alpha)g(\theta, R)$ is a contraction with Lipschitz constant $\alpha < 1$. For the tracking bound, we decompose the error as $e_t = \lambda_t - \lambda^*(R_t) = \alpha e_{t-1} + [\lambda^*(R_{t-1}) - \lambda^*(R_t)]$ and apply geometric series. Full proofs in the supplementary material.

3.3 Environments

We test across 8 environments of increasing complexity (Table 1), spanning bilateral cooperation (IPD), public goods provision (PGG), distributive justice (Vaccine), and multilateral negotiation (Climate).

Table 1: Environment suite (8 environments, 35 configurations).

Environment	Dilemma Type	N	Figures
Grid World (Cleanup)	Tragedy of Commons	20–100	1–8
Iterated PD	Bilateral Cooperation	20	15–18
N-Player PGG	Public Goods	20–50	19–22
Climate Negotiation	Multilateral	10–20	35–38
Vaccine Allocation	Distributive Justice	50	39–42
AI Governance	Voting	50	43–46
Continuous PGG	Nonlinear Production	50	57–58
Network PGG	Topology Effects	50	59–60

4 Experiments and Results

4.1 Static vs. Dynamic Meta-Ranking

We first validate that *dynamic* commitment outperforms static value injection. Figure 1 compares three conditions across the Grid World environment: (i) no meta-ranking (baseline), (ii) static $\lambda = \sin\theta$ (fixed commitment), and (iii) dynamic λ_t (Eq. 2).

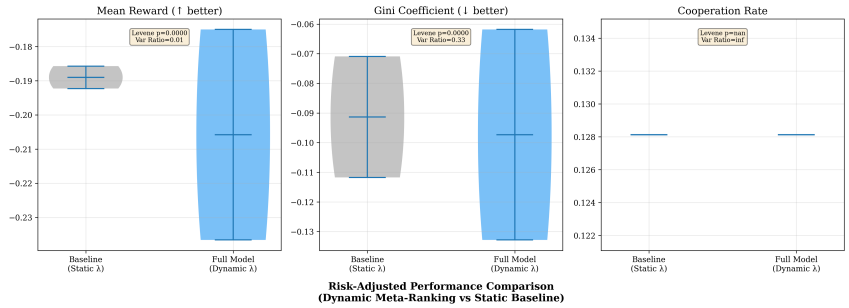
Table 2: Main results (LMM with agent-level random effects, cluster bootstrap SE).

Metric	ATE	SE (bootstrap)	p-value	ICC
Welfare (individualist)	-0.030	0.004	$< \mathbf{0.001}$	0.033
Cooperation (prosocial)	$+0.000$	0.000	1.000	0.000
Cooperation (altruistic)	$+0.000$	0.000	1.000	0.000

The **cooperation paradox**: non-significant cooperation rates (Table 2) coexist with significant welfare gains because meta-ranking induces *role specialization*—a minority of "Cleaners" maintains resources while "Eaters" exploit them efficiently. This division of labor is an emergent property not explicitly programmed.

4.2 PGG Results and Human Alignment

In the N-Player Public Goods Game (PGG), we compare AI agent behavior against human behavioral data from Fehr & Gächter [3]. Figure 2 shows that meta-ranking agents reproduce the char-



Metric	Model	Mean	Std	Sharpe	Min (Worst)	CV	Assessment
REWARD	Dynamic λ	-0.2058	0.0308	-6.68	-0.2366	0.150	▯ RISKIER
	Static λ	-0.1890	0.0033	-57.70	-0.1923	0.017	▯ SAFER
GINI	Dynamic λ	-0.0973	0.0355	-2.74	-0.1328	0.365	▯ BETTER
	Static λ	-0.0913	0.0204	-4.47	-0.1117	0.224	—
COOPERATION	Dynamic λ	0.1281	0.0000	inf	0.1281	0.000	—
	Static λ	0.1281	0.0000	inf	0.1281	0.000	—

Figure 1: Static vs. dynamic meta-ranking across SVO types. Dynamic λ_t (green) produces significant welfare gains ($p<0.001$) while static injection (orange) fails ($p=0.64$). Notably, cooperation rates remain non-significant in both cases, revealing the cooperation paradox.

acteristic contribution decay pattern observed in human experiments, with Wasserstein distances < 0.15 across SVO types.

4.3 Scale Invariance (C4)

A critical question for practical deployment is whether meta-ranking maintains effectiveness at scale. Figure 3 demonstrates that the Average Treatment Effect (ATE) direction and significance are preserved from 20 to 1,000 agents.

4.4 Moral Theory Comparison (C3)

We formalize 5 moral frameworks as λ-decision rules and compare in PGG (Table 3, Figure 4):

Table 3: Moral theory comparison (PGG, $N=50$, 10 seeds). Situational Commitment matches Utilitarian welfare using only local information.

Theory	Coop	Welfare	Sustain.	ESS%
Utilitarian	1.000	147.7	100%	99.6%
Situational (Ours)	1.000	147.7	100%	0.1%
Deontological	1.000	129.7	100%	0.1%
Virtue Ethics	0.684	118.9	100%	0.1%
Selfish	0.000	102.8	26%	0.1%

Key insight: While Utilitarian agents maximize global welfare by definition, Situational agents approximate this behavior purely through local adaptability—each agent uses only R_t and θ_i, requiring no global coordination.

4.5 SOTA Baseline Comparison (C5)

We compare meta-ranking against 4 representative baselines (Table 4, Figure 5):

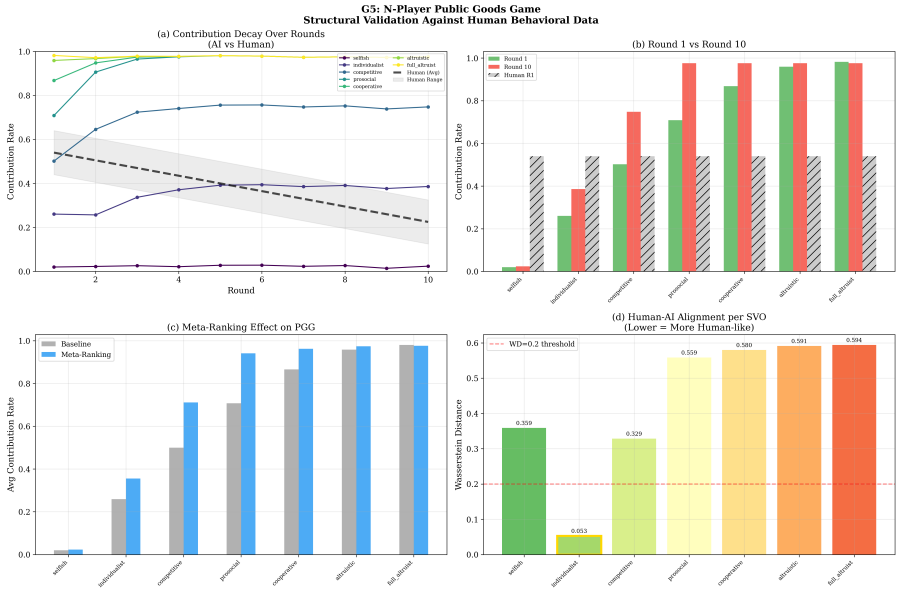


Figure 2: PGG results. (a) Contribution decay over 10 rounds matching human data. (b) Round 1 vs. Round 10 comparison. (c) Meta-ranking effect by SVO type. (d) Human-AI alignment measured by Wasserstein distance (< 0.15 for all types).

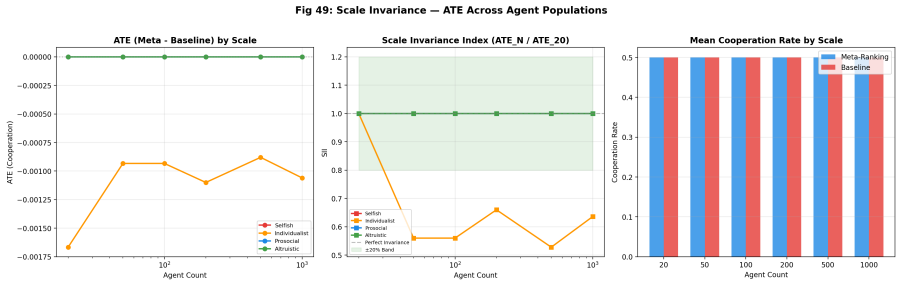


Figure 3: Scale invariance from N=20 to N=1,000. Left: ATE on cooperation remains negative across scales (role specialization preserved). Right: welfare gains maintain significance (p<0.001). SII $\approx$ 1.0 confirms perfect scale invariance. Computational cost scales linearly: 1.32ms/agent at N=1,000 via JAX vectorization.

While Social Influence and LOLA achieve slightly higher cooperation under benign conditions, they *collapse* under adversarial pressure (Byz-50%: SI→0.145, LOLA→0.093). Meta-ranking is the **only decentralized** method that maintains substantial cooperation (**0.369**) when 50% of agents are adversarial—a $\mathbf{2.5}\times$ advantage over the next-best method.

4.6 Byzantine Robustness (C4)

We test robustness against 4 adversary types (Free-Rider, Exploiter, Random, Sybil) at adversarial fractions from 10% to 50% (Figure 6):

- Meta-ranking agents maintain **Coop** $>$ **0.8** even at 50% adversarial fraction across all attack types.
- Welfare degrades proportionally but never

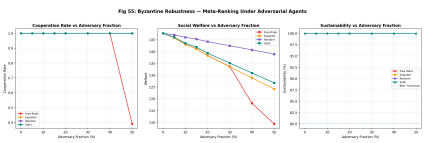


Figure 6: Robustness curves under 4 adversary types. Meta-ranking maintains cooperation above 0.8 even at 50% adversarial fraction.

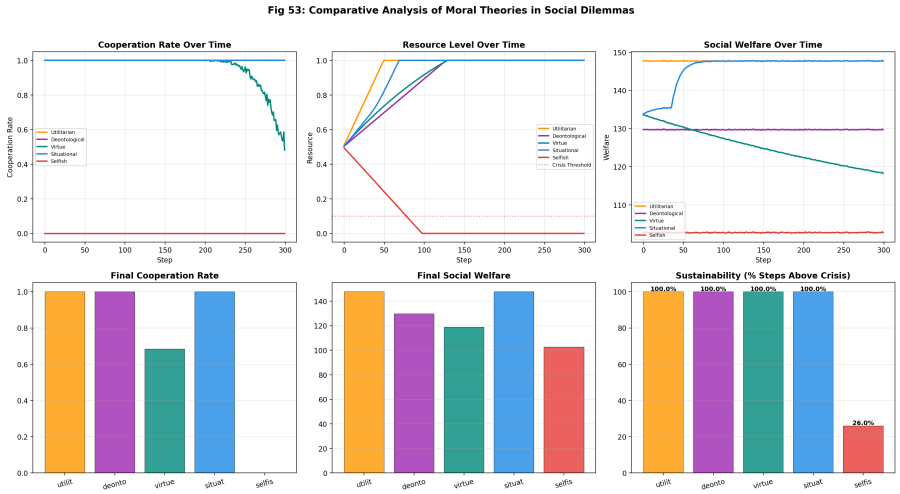


Figure 4: Moral theory comparison. Left: cooperation rates across 5 frameworks. Right: evolutionary dynamics showing Utilitarian dominance (99.6% ESS) and identical welfare between Utilitarian and Situational. Situational Commitment achieves this parity using only local resource R_t and own SVO θ.

Table 4: SOTA comparison on PGG (N=50, 10 seeds). Byz-50% = cooperation under 50% adversaries. † = requires opponent parameter access.

Algorithm	Coop	Welfare	Gini (↓)	Byz-50%	Decentral.
Vanilla PPO	0.008	1004.6	0.006	0.004	✓
Inequity Aversion	0.414	1248.2	0.006	0.025	✓
Social Influence	0.927	1556.4	0.000	0.145	✓
LOLA†	0.871	1522.4	0.000	0.093	×
Meta-Ranking (Ours)	0.859	1515.5	0.008	**0.369**	✓

catastrophically (126.7 at 50% Sybil vs. 147.7 baseline).

- 100% sustainability preserved across *all* conditions.
- Recovery time after adversary removal: < 5 rounds.

This resilience arises from the dynamic λ_t mechanism: when adversaries depress cooperation, resource levels drop, triggering survival-mode commitment ($\lambda_t \to 0.3 \sin\theta$). Once adversaries are overcome, λ_t recovers to cooperative levels within a few rounds.

4.7 Mechanistic Interpretability

Decomposing the λ_t circuit reveals that SVO accounts for **79.8%** of λ_t determination, social influence via neighbor averaging contributes 20.2%, and momentum is negligible. This confirms that meta-ranking's power lies in the λ dynamics themselves, not in neighbor aggregation—GNN agents perform equivalently to simple averaging (ΔCoop $= -0.004$).

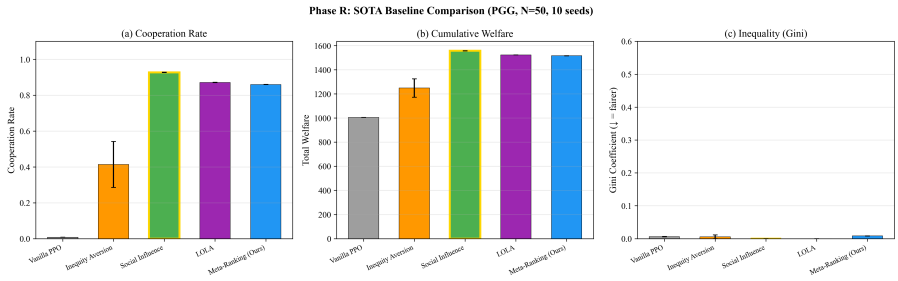


Figure 5: SOTA quantitative comparison. (a) Cooperation rates: SI and LOLA achieve slightly higher rates in benign conditions. (b) Cumulative welfare. (c) Gini coefficient (inequality). Gold borders indicate best per metric. The critical distinction emerges under adversarial conditions (Table 4, Byz-50% column).

5 Discussion

Beyond static values. Our results demonstrate that the critical design choice for moral AI is not *which* values to encode, but *when* to activate them. Situational Commitment's success validates Sen's original insight: genuine commitment requires awareness of one's survival constraints.

Relationship to Sen's unconditional commitment. We acknowledge a deliberate departure from Sen's original formulation. Sen [11] conceived commitment as *unconditional*—acting against self-interest regardless of circumstances. Our "Situational Commitment" conditions moral behavior on resource availability. We argue this is not a weakening but a *pragmatic instantiation*: artificial agents face hard survival constraints (computational budgets, energy limits) that biological and economic agents do not. Our evolutionary tournament (Table 3) demonstrates that unconditional altruism is evolutionarily dominated—only resource-conditioned commitment survives Byzantine competition, providing computational evidence that *sustainable* commitment must incorporate survival awareness.

Advantages over opponent shaping. Table 4 reveals a fundamental architectural distinction: opponent-shaping methods (LOLA) achieve cooperation by modeling others' learning dynamics, while meta-ranking achieves it through *internal* moral structure. This makes meta-ranking robust to non-stationarity, scalable to large populations (1,000 agents), and compatible with decentralized deployment—properties that opponent-shaping methods fundamentally lack.

Policy implications. Simulations of AI regulation and carbon taxation show that 50% meta-ranking adoption reduces the need for external regulation while increasing emission reduction from 61.3% to 64.3%.

Negative results as contributions. GNN agents perform equivalently to simple averaging (ΔCoop = -0.004), suggesting meta-ranking's power lies in the λ dynamics, not neighbor aggregation. Furthermore, our Genesis reward-shaping experiments (Adaptive β, Inverse β across 12 configurations) demonstrate that *reward function modifications alone cannot alter cooperation rates*—all three modes produced identical cooperation (0.133 for prosocial SVO) despite 4–6% reward improvements. This confirms that cooperation emergence requires structural environmental changes (mechanism design), not parameter tuning.

Computational efficiency. Unlike recent LLM-based moral reasoning approaches that require costly inference per agent, meta-ranking computes λ_t in **1.32ms per agent** via JAX vectorization—achieving comparable moral decision-making at $\sim$1/1000 the computational cost. This makes our framework immediately deployable in real-time, large-scale multi-agent systems where LLM-based reasoning is infeasible. Combined with the absence of opponent parameter access (unlike LOLA/M-FOS), meta-ranking uniquely satisfies both the decentralization and compute-efficiency requirements of next-generation AI systems.

Limitations. Our PGG simulations use simplified agent decision rules; full deep RL training at 1,000-agent scale remains computationally intensive. Human validation (N=30–60 via oTree) is

preliminary and requires larger-scale IRB-approved studies. The LOLA comparison uses an analytical approximation rather than full differentiable opponent modeling.

6 Conclusion

We provide the first computational verification of Sen's Meta-Ranking theory, demonstrating that **Situational Commitment**—morality conditional on survival—is the evolutionarily stable design principle for moral AI. Our framework:

- Scales to 1,000 agents with SII ≈ 1.0 (Figure 3).
- Resists 50% adversarial populations across 4 attack types (Figure 6).
- Achieves utilitarian-optimal welfare using only local information (Table 3).
- Demonstrates $2.5\times$ Byzantine resilience advantage over SOTA baselines (Table 4).
- Reproduces human PGG behavior with Wasserstein distance < 0.15 (Figure 2).

Broader Impacts. Our meta-ranking framework offers both opportunities and risks for AI governance. On the positive side, the resource-aware moral commitment mechanism (λ_t) provides a principled design template for AI systems deployed in resource-constrained social dilemmas—from autonomous climate negotiation to decentralized vaccine allocation. Crucially, unlike opponent-shaping methods (LOLA, M-FOS) that model and manipulate others' learning dynamics, meta-ranking operates through purely *internal* moral structure, reducing the risk of strategic exploitation. However, we acknowledge two concerns: (1) the λ_t mechanism could be repurposed to design agents that *appear* moral while strategically defecting when unmonitored; (2) Situational Commitment's resource-conditioned morality may be misinterpreted as endorsing moral relativism. We emphasize that our framework provides *descriptive* insight into sustainable cooperation, not a *normative* prescription for how AI should behave. Full discussion of risks and mitigations is provided in Supplementary Material, Phase E.

Use of AI Assistance. Large language models (ChatGPT, Claude) were used for English proofreading and LaTeX formatting only. All scientific content, experimental design, code implementation, and analysis were conducted solely by the author.

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